Tislelizumab plus chemotherapy versus placebo plus chemotherapy as first-line treatment for advanced or metastatic oesophageal squamous cell carcinoma (RATIONALE-306): a global, randomised, placebo-controlled, phase 3 study

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Summary

Background The options for first-line treatment of advanced oesophageal squamous cell carcinoma are scarce, and the outcomes remain poor. The anti-PD-1 antibody, tislelizumab, has shown antitumour activity in previously treated patients with advanced oesophageal squamous cell carcinoma. We report interim analysis results from the RATIONALE-306 study, which aimed to assess tislelizumab plus chemotherapy versus placebo plus chemotherapy as first-line treatment for advanced or metastatic oesophageal squamous cell carcinoma.

Methods This global, randomised, double-blind, parallel-arm, placebo-controlled, phase 3 study was conducted at 162 medical centres across Asia, Europe, Oceania, and North America. Patients (aged ≥18 years) with unresectable, locally advanced, recurrent or metastatic oesophageal squamous cell carcinoma (regardless of PD-L1 expression), Eastern Cooperative Oncology Group performance status of 0–1, and measurable or evaluable disease per Response Evaluation Criteria in Solid Tumours (version 1.1) were recruited. Patients were randomly assigned (1:1), using permuted block randomisation (block size of four) and stratified by investigator-chosen chemotherapy, region, and previous definitive therapy, to tislelizumab 200 mg or placebo intravenously every 3 weeks on day 1, together with an investigator-chosen chemotherapy doublet, comprising a platinum agent (cisplatin 60–80 mg/m² intravenously on day 1 or oxaliplatin 130 mg/m² intravenously on day 1) plus a fluoropyrimidine (fluorouracil [750–800 mg/m² intravenously on days 1–5] or capecitabine [1000 mg/m² orally twice daily on days 1–14]) or paclitaxel (175 mg/m² intravenously on day 1). Treatment was continued until disease progression or unacceptable toxicity. Investigators, patients, and sponsor staff or designees were masked to treatment. The primary endpoint was overall survival. The efficacy analysis was done in the intention-to-treat population (ie, all randomly assigned patients) and safety was assessed in all patients who received at least one dose of study treatment. The trial is registered with ClinicalTrials.gov, NCT03783442.

Findings Between Dec 12, 2018, and Nov 24, 2020, 869 patients were screened, of whom 649 were randomly assigned to tislelizumab plus chemotherapy (n=326) or placebo plus chemotherapy (n=323). Median age was 64·0 years (IQR 59·0–69·0), 563 (87%) of 649 participants were male, 86 (13%) were female, 486 (75%) were Asian, and 155 (24%) were White. 324 (99%) of 326 patients in the tislelizumab group and 321 (99%) of 323 in the placebo group received at least one dose of the study drug. As of data cutoff (Feb 28, 2022), median follow-up was 16·3 months (IQR 8·6–21·8) in the tislelizumab group and 9·8 months (IQR 5·8–19·0) in the placebo group, and 196 (60%) of 326 patients in the tislelizumab group versus 226 (70%) of 323 in the placebo group had died. Median overall survival in the tislelizumab group was 17·2 months (95% CI 15·8–20·1) and in the placebo group was 10·6 months (9·3–12·1; stratified hazard ratio 0·66 [95% CI 0·54–0·80]; one-sided p<0·0001). 313 (97%) of 324 patients in the tislelizumab group and 309 (96%) of 321 in the placebo group had treatment-related treatment-emergent adverse events. The most common grade 3 or 4 treatment-related treatment-emergent adverse events were decreased neutrophil count (99 [31%] in the tislelizumab group *vs* 105 [33%] in the placebo group), decreased white blood cell count (35 [11%] *vs* 50 [16%]), and anaemia (47 [15%] *vs* 41 [13%]). Six deaths in the tislelizumab group (gastrointestinal and upper gastrointestinal haemorrhage [n=2], myocarditis [n=1], pulmonary tuberculosis [n=1], electrolyte imbalance [n=1], and respiratory failure [n=1]) and four deaths in the placebo group (pneumonia [n=1], septic shock [n=1], and unspecified death [n=2]) were determined to be treatment-related.

Interpretation Tislelizumab plus chemotherapy as a first-line treatment for advanced or metastatic oesophageal squamous cell carcinoma provided superior overall survival with a manageable safety profile versus placebo plus chemotherapy. Given that the interim analysis met its superiority boundary for the primary endpoint, as confirmed by the independent data monitoring committee, this Article represents the primary study analysis.

Lancet Oncol 2023; 24: 483–95

Published **Online** April 17, 2023 https://doi.org/10.1016/S1470-2045(23)00108-0

This online publication has been corrected. The corrected version first appeared at thelancet.com/oncology on February 26, 2024

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Funding BeiGene.

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Introduction

Oesophageal cancer is the tenth most common cancer worldwide and sixth most prevalent cause of death due to cancer.[1] More than two-thirds of patients with oesophageal cancer are diagnosed with advanced or metastatic disease; 5-year survival is 26% for disease with regional lymph node metastases and 5% with distant metastases.[2,3] Oesophageal squamous cell carcinoma comprises approximately 90% of all oesophageal cancers.[4] Recommended first-line treatment for patients with advanced or metastatic oesophageal squamous cell carcinoma is typically a platinum-based (cisplatin or oxaliplatin) regimen with a fluoropyrimidine or paclitaxel;[4,5] however, the benefit achieved has generally been unsatisfactory, with median overall survival rarely surpassing 10 months.[5,6]

Anti-PD-1 antibodies given with chemotherapy have superior overall survival benefit versus chemotherapy as a first-line treatment for oesophageal squamous cell carcinoma.[7–11] However, existing clinical evidence in global populations is mainly based on studies combining anti-PD-1 monoclonal antibodies with one chemotherapy doublet option, cisplatin plus fluorouracil.[7,8] This fact is concerning because increasing evidence indicates potential differences between classes of cytotoxic agents in their ability to induce immunogenic cell death and cooperate with immunotherapy.[12–16] Phase 3 clinical evidence for other chemotherapy doublet options (eg, cisplatin plus paclitaxel) in oesophageal squamous cell carcinoma is limited to studies done almost entirely in Asia,[9–11] the findings of which might not be representative of the wider global population because of

Research in context

Evidence before this study

We searched PubMed from database inception to Dec 12, 2018 (the start date of the present RATIONALE-306 trial) for published reports of clinical trial results using the search terms "PD-1" OR "PD-L1" and "esophageal cancer" or "oesophageal cancer" (and related terms) in the title or abstract without language restrictions. From this search, we identified reports from the KEYNOTE-180 (pembrolizumab) and ATTRACTION-1 (nivolumab) studies, which indicated that anti-PD-1 treatment as second-line or later therapy for advanced or metastatic oesophageal squamous cell carcinoma provided durable antitumour activity and a manageable safety profile. However, we found no publications supporting anti-PD-1-based therapy in the first-line setting, and chemotherapy remained the standard first-line treatment option for advanced oesophageal squamous cell carcinoma, despite offering only short-term overall survival benefit. We did a subsequent search of PubMed for first-line studies published from database inception to June 30, 2022, and identified results from the global, phase 3 KEYNOTE-590 and CheckMate 648 trials that found a significant overall survival benefit for pembrolizumab plus chemotherapy and nivolumab plus chemotherapy, respectively, as first-line treatments for advanced oesophageal squamous cell carcinoma. These results led to the incorporation of anti-PD-1 therapies as part of recommended first-line regimens in this setting. However, these studies only investigated the combination of anti-PD-1 therapy with one chemotherapy doublet option, cisplatin and fluorouracil. Phase 3 clinical evidence for other doublet options (eg, cisplatin plus paclitaxel) combined with anti-PD-1 therapies in oesophageal squamous cell carcinoma was limited to studies done almost entirely in Asia (ie, ORIENT-15, ESCORT-1st, JUPITER-06), which might not be representative of the wider, global population.

Added value of this study

To our knowledge, RATIONALE-306 is the first international study designed to investigate the effects of an anti-PD-1 monoclonal antibody in combination with different chemotherapy regimens—oxaliplatin or cisplatin plus fluoropyrimidine (capecitabine or fluorouracil) or paclitaxel—at the investigators' discretion and as recommended by international, local, or country-specific treatment guidelines, thereby closely aligning with real-world medical practice across multiple regions. Increasing evidence suggests that substantial diversity exists among the immunostimulatory or regulatory properties of cytotoxic agents, and no global consensus exists regarding optimal first-line chemotherapy. In RATIONALE-306, we observed overall survival benefit when adding tislelizumab to chemotherapy across different chemotherapy options.

Implications of all the available evidence

Clinical practice guidelines for oesophageal cancer were updated in 2021 to include anti-PD-1 monoclonal antibody plus chemotherapy combinations as first-line treatment options in advanced oesophageal squamous cell carcinoma and include recommendations for pembrolizumab plus chemotherapy (particularly for tumours with high PD-L1 expression) and nivolumab plus chemotherapy (particularly for tumours with high PD-L1 expression). The results from the RATIONALE-306 study indicate that tislelizumab plus chemotherapy provides an effective first-line treatment for advanced oesophageal squamous cell carcinoma and supports the addition of this combination to the treatment options for advanced oesophageal squamous cell carcinoma.

differences in epidemiology, genomic profiles, practice preference, and prognosis between Asian and other patient populations worldwide.[2,8,17–19]

Tislelizumab is a humanised IgG4 monoclonal antibody with high affinity and binding specificity for PD-1.[20,21] In patients with advanced or metastatic oesophageal squamous cell carcinoma who had disease progression after first-line systemic chemotherapy, tislelizumab monotherapy showed superior overall survival compared with chemotherapy in the global, phase 3 RATIONALE-302 study.[22,23] Here, we report results from the RATIONALE-306 study, which assessed efficacy and safety of first-line treatment with tislelizumab plus investigator-chosen chemotherapy (platinum plus fluoropyrimidine or paclitaxel) compared with placebo plus investigator-chosen chemotherapy in patients with advanced or metastatic oesophageal squamous cell carcinoma.

Methods

Study design and participants

RATIONALE-306 is a global, randomised, double-blind, parallel-arm, placebo-controlled, phase 3 study. Patients were recruited from 162 medical sites across Asia, Europe, North America, and Oceania (appendix pp 2–14). Patients aged 18 years or older with histologically confirmed, unresectable, locally advanced, recurrent or metastatic oesophageal squamous cell carcinoma not treated with previous systemic therapy for advanced disease, and not amenable to definitive therapies (including surgery or radiation) per local investigator, were recruited. If patients had received previous neoadjuvant or adjuvant platinum-based chemotherapy, a treatment-free interval of at least 6 months was required. Patients were required to have an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1, and measurable or evaluable disease per Response Evaluation Criteria in Solid Tumours (RECIST; version 1.1). Patients were enrolled regardless of expression status of PD-L1. Key exclusion criteria were brain or leptomeningeal metastases that were symptomatic or required treatment, evidence of complete oesophageal obstruction not amenable to treatment, evidence of fistula, active autoimmune diseases, medical conditions requiring systemic corticosteroids or immunosuppressants, or previous therapies targeting PD-1, PD-L1, or PD-L2. Full eligibility criteria are in the appendix (pp 15–17).

Approval was obtained from the relevant institutional review board or independent ethics committee for each study site. The study was conducted in accordance with the International Conference on Harmonisation Good Clinical Practice Guideline, the principles of the Declaration of Helsinki, and local laws and regulations. All patients provided written informed consent. Safety was monitored by an independent data monitoring committee, which also oversaw the interim overall survival analysis and made recommendations on the basis of predefined criteria for overall survival, and results from the objective response rate and other secondary efficacy endpoints. The full protocol is available in the appendix.

Randomisation and masking

Patients were randomly assigned (1:1), using an interactive response technology system with permuted block randomisation (block size of four), to either tislelizumab plus investigator-chosen chemotherapy or placebo plus investigator-chosen chemotherapy. Randomisation was stratified by investigator-chosen chemotherapy (platinum plus fluoropyrimidine *vs* platinum plus paclitaxel), region (Asia [excluding Japan] *vs* Japan *vs* other regions), and previous definitive therapy (yes *vs* no). Investigators, patients, and sponsor staff or its designees were masked to group assignment. After achieving the primary objective, the study was to be unblinded.

Procedures

Patients were given tislelizumab 200 mg or matching placebo intravenously every 3 weeks on day 1 of 21-day cycles, together with an investigator-chosen chemotherapy doublet. Chemotherapy options were a platinum agent (cisplatin 60–80 mg/m² intravenously on day 1 or oxaliplatin 130 mg/m² intravenously on day 1) combined with a fluoropyrimidine (fluorouracil [750–800 mg/m² intravenously on days 1–5] or capecitabine [1000 mg/m² orally twice daily on days 1–14]) or paclitaxel (175 mg/m² intravenously on day 1). Cisplatin was used as the platinum agent in China, Taiwan, and Japan (oxaliplatin substitution was not permitted per local treatment guidelines).

See Online for appendix

Treatment with the assigned study drugs was continued until investigator-assessed disease progression per RECIST version 1.1, unacceptable toxicity, death, or withdrawal of consent. Cisplatin or oxaliplatin could be stopped after six cycles per site or investigator preference, or per standard practice by country. After stopping platinum agent, fluoropyrimidine or paclitaxel could continue at the regular schedule, if appropriate. After 2 years of study treatment, if complete response, partial response, or stable disease was reached, tislelizumab or placebo treatment could be stopped on the basis of investigator's evaluation, considering the patient's clinical benefit and risk. Cross-over between treatment groups or between fluoropyrimidine and paclitaxel during the study treatment period was prohibited, even after unmasking. After treatment discontinuation, subsequent anticancer therapy was to be determined by investigators according to local treatment guidelines and medical practice.

Demographic details of the patient's sex and race were self-reported. Assessments of radiological tumour response by CT or MRI were done every 6 weeks for the first 48 weeks, then every 9 weeks thereafter locally by investigator and centrally by blinded independent review committee (BIRC) per RECIST version 1.1, until disease progression or death. Confirmation of tumour response

was not required per protocol. Patients could continue study treatment after disease progression if they were benefiting from study treatment, as assessed by investigators. Adverse event assessments were required for all scheduled and unscheduled visits, which were conducted face-to-face at the site (recommended), by telephone, or via other means (eg, review of medical records if the patient was unable to attend an onsite visit), throughout the study period and up to 30 days after the last dose of study treatment (including chemotherapy) or initiation of new anticancer therapy, whichever occurred first. Adverse events were graded per National Cancer Institute Common Terminology Criteria for Adverse Events (version 4.03) and safety was assessed by monitoring treatment-emergent adverse events and serious adverse events, mapped to the Medical Dictionary for Regulatory Activities terms. Treatment-related adverse events included treatment-emergent adverse events that were considered by the investigator to be related to the study drug or treatment-emergent adverse events with a missing causality. Immune-related adverse events were reported until 90 days after the last dose of tislelizumab or placebo, regardless of initiation of new anticancer therapy (to be reported elsewhere).

PD-L1 score was centrally assessed retrospectively with all available tumour samples by tumour area positivity (TAP) score, defined as total percentage of tumour area (tumour and any desmoplastic stroma) covered by tumour cells with PD-L1 membrane staining at any intensity and tumour-associated immune cells with PD-L1 staining at any intensity, as visually estimated by pathologists using the VENTANA PD-L1 (SP263) assay (742-4907, Ventana Medical Systems, Tucson, AZ, USA; PD-L1 TAP algorithm is described briefly in the appendix [p 18]).[24,25] Pathologists in the central laboratory (Q Squared Solutions, Edinburgh, UK and Q Squared Solutions [Beijing], Beijing, China) scored the same stained samples according to combined positive score (number of PD-L1-expressing tumour cells, lymphocytes, and macrophages divided by total number of viable tumour cells, multiplied by 100) and tumour cell score (percentage of viable tumour cells with any membrane staining above background) for exploratory purposes.

Outcomes

The primary endpoint was overall survival, defined as time from randomisation until death from any cause. Secondary endpoints were progression-free survival, defined as time from randomisation to first documentation of investigator-assessed disease progression per RECIST version 1.1 or death, whichever occurred first; overall response rate, defined as the proportion of patients whose best overall response was complete response or partial response assessed by the investigator per RECIST version 1.1; overall survival in the subgroup with a PD-L1 TAP score of 10% or more; duration of response, defined as time from first determination of objective response until first documentation of investigator-assessed progression per RECIST version 1.1 or death, whichever occurred first; safety (incidence and severity of adverse events); and health-related quality of life, which will be reported in a separate publication. Exploratory endpoints included investigator-assessed disease control rate (proportion of patients whose best overall response was complete response, partial response, or stable disease, per RECIST version 1.1) and blinded independent review committee-assessed progression-free survival, overall response rate, duration of response, and disease control rate. A complete list of exploratory endpoints is listed in the appendix (p 19) and in the protocol.

Statistical analysis

Efficacy analyses were done in the intention-to-treat (ITT) analysis set, which included all patients randomly assigned to treatment. Safety was assessed in all patients who received at least one dose of study treatment (safety population). For assessment of overall response rate, we included all patients in the ITT population and those without a post-baseline tumour assessment before data cutoff (for any reason, including death) or start of new anticancer treatment were considered as not evaluable for best overall response and therefore as non-responders for overall response rate.

For the final analysis, approximately 488 events were required to provide 90% power at a one-sided significance level of 0·025 to detect superiority of tislelizumab plus chemotherapy versus placebo plus chemotherapy in overall survival. The hazard ratio (HR) for the primary endpoint of overall survival in the ITT population for tislelizumab plus chemotherapy versus placebo plus chemotherapy was assumed to be 0·74 at the time of final analysis, corresponding to a median overall survival of 12·0 months with tislelizumab plus chemotherapy versus 9·0 months with placebo plus chemotherapy. An interim analysis was planned when approximately 423 deaths were reached, corresponding to 87% of the target number of deaths for the final analysis and an estimated power of 81%. The superiority boundary at the interim overall survival analysis was predefined using the O'Brien-Fleming boundary approximated using the Hwang-Shih-DeCani spending function, with a gamma parameter set at –4, and could be adjusted on the basis of observed number of deaths at the time of interim analysis. Based on 422 actual observed deaths at the time of the interim analysis, the updated one-sided p value boundary for superiority was 0·0144. The independent data monitoring committee confirmed that the study met the specified one-sided p value boundary for superiority of the primary endpoint after reviewing the results of the interim analysis; the interim analysis was then regarded as the primary study analysis. The originally intended final analysis will not be pursued.

If the null hypothesis for overall survival was rejected, the corresponding α would be shifted to the hypothesis

tests of the hierarchically tested secondary endpoints: progression-free survival by investigator assessment, overall response rate by investigator assessment, overall survival in patients with a PD-L1 TAP score of at least 10%, and health-related quality of life, which would be tested sequentially; further details are in the statistical analysis plan (appendix). A one-sided p value of less than 0·025 was considered to be significant for secondary endpoints under the hierarchical testing.

The null hypothesis for overall survival and progression-free survival was tested using a one-sided p value, estimated from a log-rank test, stratified by investigator-chosen chemotherapy (platinum with fluoropyrimidine *vs* platinum with paclitaxel), pooled geographical region (Asia *vs* other regions outside Asia), and previous definitive therapy (yes *vs* no). HRs and two-sided 95% CIs were estimated with a Cox regression model including treatment as a covariate, with the same stratification factors as for the null hypothesis for overall survival and progression-free survival (prespecified in the protocol and statistical analysis plan). The Kaplan-Meier method was used to estimate median overall survival (in the ITT population and in each subgroup with PD-L1 TAP score [<10% *vs* ≥10%]) and progression-free survival. For median overall survival and progression-free survival, two-sided 95% CIs were constructed using a generalised Brookmeyer and Crowley method.

The primary endpoint of overall survival was also assessed in prespecified subgroups by age (<65 *vs* ≥65 years), sex (male *vs* female), smoking status (former or current smoker *vs* never smoker), PD-L1 expression status using TAP score (<10% *vs* ≥10% *vs* unknown), investigator-chosen chemotherapy (platinum plus fluoropyrimidine *vs* platinum plus paclitaxel), ECOG performance status (0 *vs* 1), geographical region (Asia *vs* other regions and Asia excluding Japan *vs* Japan *vs* other regions), race (Asian and other races *vs* White), disease status at entry (metastatic *vs* locally advanced), and previous definitive therapy (yes *vs* no). Additionally, overall survival was assessed in post-hoc patient subgroups by choice of chemotherapy doublet regimen and PD-L1 expression status using combined positive score and tumour cell score. The interaction effect of those subgroup factors including PD-L1 expression, scored on the basis of TAP, combined positive score and tumour cell score on treatment was assessed as a post-hoc analysis using an unstratified Cox regression model including treatment, each subgroup factor, and their interaction term as covariates. $p_{interaction}$ values greater than or equal to 0·15 were considered non-significant. Subgroup analyses for progression-free survival included prespecified analyses for geographical region (Asia *vs* other regions) and PD-L1 expression status (TAP score <10% *vs* ≥10% *vs* unknown) and post-hoc analyses for investigator-chosen chemotherapy (platinum plus fluoropyrimidine *vs* platinum plus paclitaxel). A prespecified multivariable analysis was also conducted for overall survival, adjusting for key baseline characteristics and prognostic factors based on a stratified Cox regression model, including treatment group, baseline PD-L1 TAP score, age, sex, smoking status, ECOG performance status, and disease stage as covariates, and pooled geographical region, previous definitive therapy, and investigator-chosen chemotherapy as strata.

Overall response rate was tested using the Cochran-Mantel-Haenszel test, adjusting for prespecified stratification factors; the two-sided 95% CI for odds ratio was calculated alongside Clopper-Pearson 95% CIs of overall response rate in each treatment group. Prespecified subgroup analyses were conducted for overall response rate as per the progression-free survival analyses. Duration of response was calculated in a similar

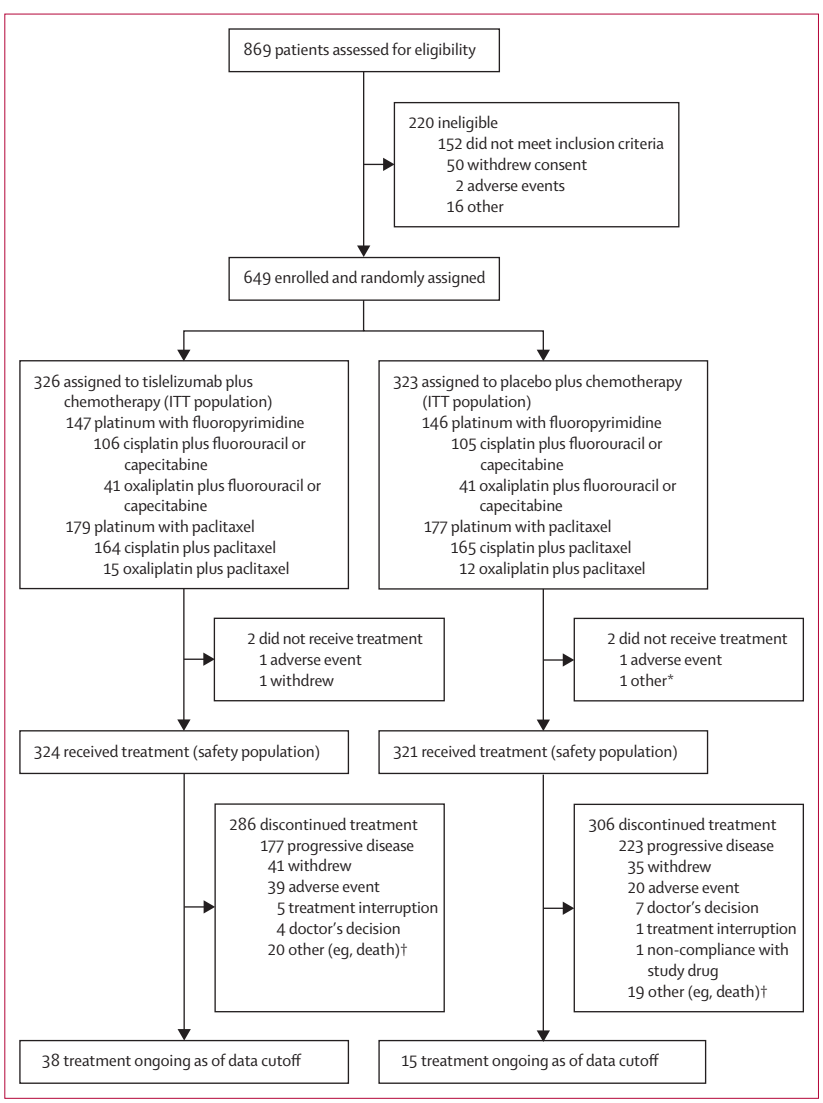

Figure 1: **Trial profile**
Data cutoff was Feb 28, 2022. ITT=intention-to-treat. *One patient not treated due to an interactive web response system technical issue. †Death was the main (but not only) reason for treatment discontinuation.

	Tislelizumab plus chemotherapy group (n=326)	Placebo plus chemotherapy group (n=323)
Age, years		
Median	64·0 (59·0–68·0)	65·0 (58·0–70·0)
<65	176 (54%)	161 (50%)
≥65	150 (46%)	162 (50%)
Sex		
Male	282 (87%)	281 (87%)
Female	44 (13%)	42 (13%)
Geographical region		
Asia*	243 (75%)	243 (75%)
Europe	79 (24%)	77 (24%)
North America	1 (<1%)	1 (<1%)
Oceania	3 (1%)	2 (1%)
Race		
Asian	243 (75%)	243 (75%)
White	79 (24%)	76 (24%)
American Indian or Alaska Native	0 (0%)	1 (<1%)
Not reported or unknown	4 (1%)	3 (1%)
BMI, kg/m²	21·2 (19·4–23·4)	21·2 (18·9–24·1)
ECOG performance status		
0	109 (33%)	104 (32%)
1	217 (67%)	219 (68%)
Smoking status		
Never	68 (21%)	81 (25%)
Current or former	247 (76%)	231 (72%)
Missing	11 (3%)	11 (3%)
Disease status at study entry		
Locally advanced	47 (14%)	41 (13%)
Metastatic	279 (86%)	282 (87%)
Number of metastatic sites at study entry		
0	47 (14%)	41 (13%)
1	144 (44%)	143 (44%)
2	81 (25%)	80 (25%)
>2	54 (17%)	59 (18%)

(Table 1 continues in next column)

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	Tislelizumab plus chemotherapy group (n=326)	Placebo plus chemotherapy group (n=323)
Histological type		
Squamous cell carcinoma	325 (>99%)	323 (100%)
Other†	1 (<1%)	0
Previous definitive therapy		
Definitive surgery‡	107 (33%)	107 (33%)
Definitive radiotherapy§	40 (12%)	40 (12%)
Definitive surgery and radiotherapy¶	4 (1%)	6 (2%)
No previous definitive therapy	183 (56%)	182 (56%)
PD-L1 expression‖		
TAP score ≥10%	116 (36%)	107 (33%)
TAP score <10%	151 (46%)	168 (52%)
Unknown**	59 (18%)	48 (15%)

Data are median (IQR) or n (%). ECOG=Eastern Cooperative Oncology Group. TAP=tumour area positivity. *Including 33 patients from Japan per group. †One patient had neuroendocrine tumour histology. ‡Definitive surgery included surgery with or without adjuvant or neoadjuvant treatment; four patients in the tislelizumab group and six in the placebo group who had received both definitive surgery and radiotherapy were included in this category. §Definitive radiotherapy included radiotherapy with or without chemotherapy; four patients in the tislelizumab group and six in the placebo group who had received both definitive surgery and radiotherapy were included in this category. ¶Patients who received previous definitive surgery and radiotherapy were included in the previous definitive therapy subgroup for all subgroup analyses. ‖PD-L1 expression by combined positive score and tumour cell score is provided in the appendix (p 22). **Unknown refers to patients without sample collection, with non-evaluable samples, or with scored unqualified samples (patients with scored unqualified samples were identified and reclassified as unknown after database lock; for details refer to appendix p 21).

Table 1: **Baseline demographic and clinical characteristics (intention-to-treat population)**

way to progression-free survival; medians were also calculated. Censoring rules for progression-free survival and duration of response are provided in the statistical analysis plan (appendix). Safety data were analysed using descriptive statistics.

All calculations and analyses were done using SAS (version 9.4 or higher). This study is registered with ClinicalTrials.gov (NCT03783442).

Role of the funding source

The funder of the study had a role in study design, data collection, data analysis, data interpretation, and writing of the clinical study report. Medical writing support was provided by the funder.

Results

Between Dec 12, 2018, and Nov 24, 2020, 869 patients were screened, of whom 649 were randomly assigned to receive tislelizumab plus chemotherapy (n=326) or placebo plus chemotherapy (n=323; ITT population; figure 1). 324 (99%) of 326 patients in the tislelizumab group and 321 (99%) of 323 in the placebo group received at least one dose of the study drug (safety population). Across all 645 patients who received at least one dose of study drug, 293 (45%) were treated with platinum plus fluoropyrimidine and 356 (55%) with platinum plus paclitaxel (figure 1). Most patients in Asia (464 [96%] of 484) received a cisplatin-based doublet, and most patients in other regions (89 [55%] of 161) received an oxaliplatin-based doublet (appendix p 20). Baseline characteristics and demographics were generally balanced between treatment groups (table 1). Median age was 64·0 years (IQR 59·0–69·0), 563 (87%) of 649 patients were male, 86 (13%) were female, 486 (75%) were Asian, 155 (24%) were White, and eight (1%) were other races.

As of data cutoff (Feb 28, 2022), median study follow-up (from randomisation to data cutoff, death, or study

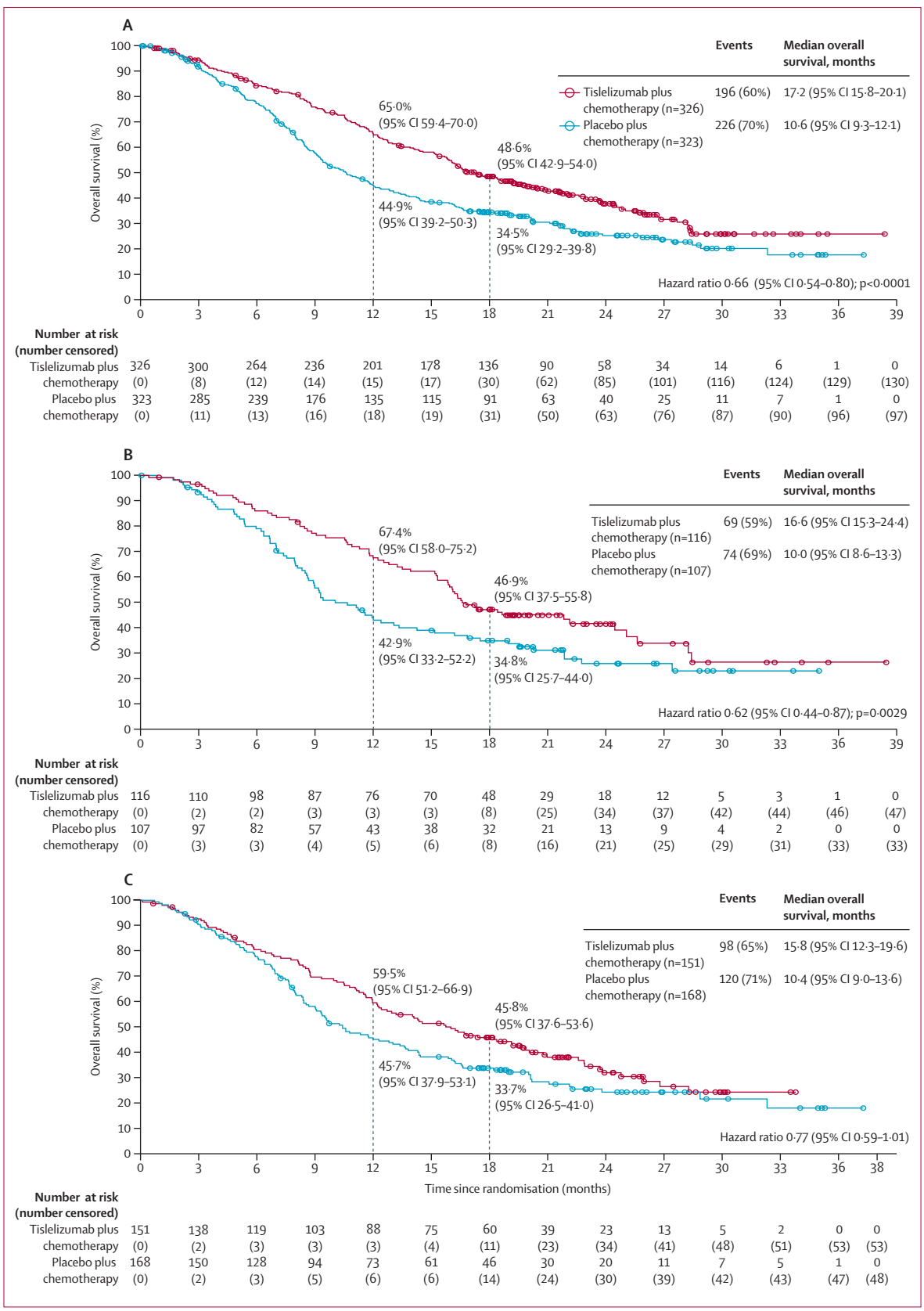

Figure 2: **Kaplan-Meier plot of overall survival in the ITT population (A) and in patients with a PD-L1 TAP score of ≥10% (B) and <10% (C)**
Overall survival data for the subgroup of patients with unknown PD-L1 TAP score at baseline are in the appendix (p 24). ITT=intention to treat. TAP=tumour area positivity.

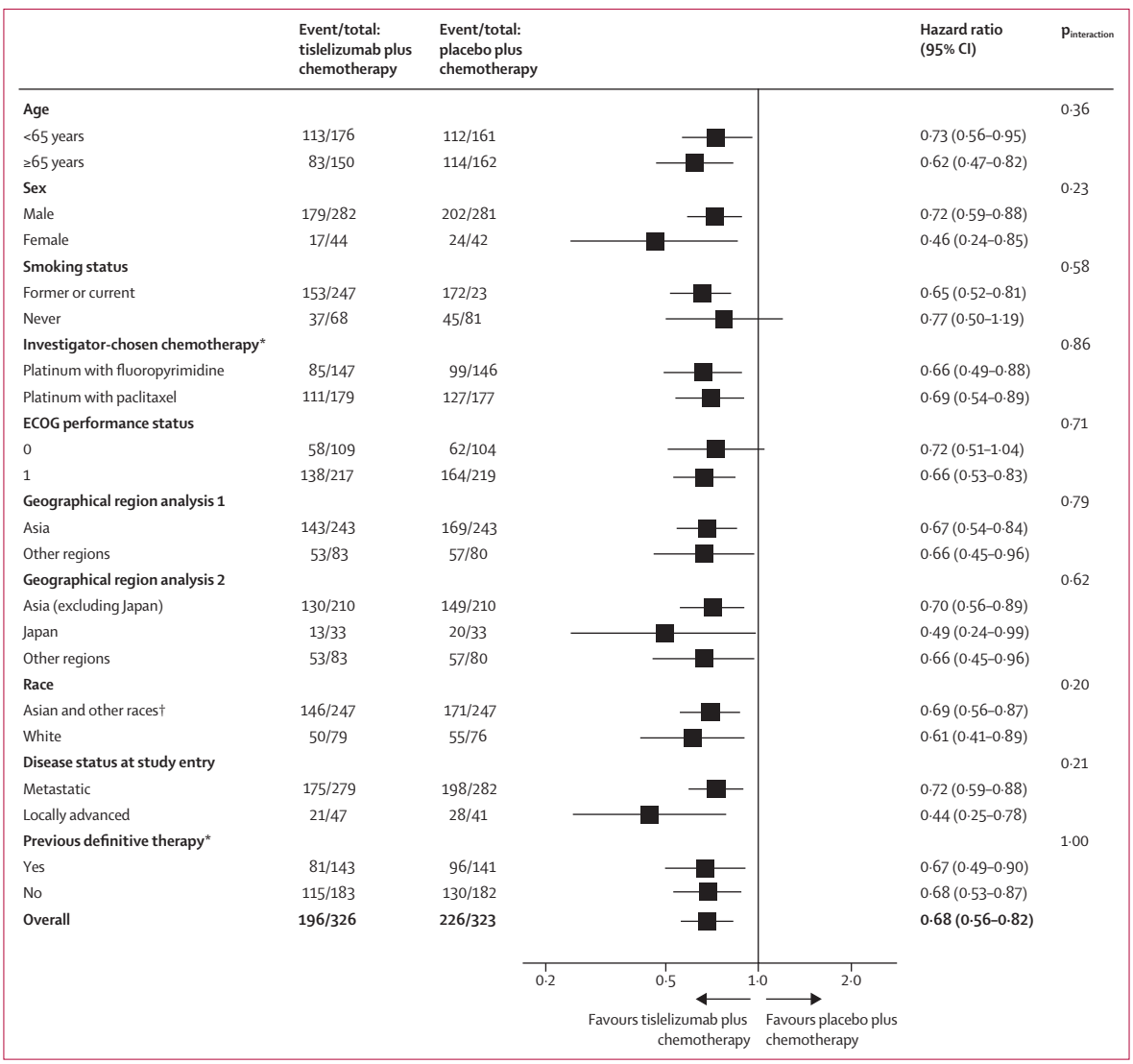

Figure 3: **Prespecified subgroup analyses of overall survival**

$p_{interaction}$ value was based on a post-hoc analysis using an unstratified Cox regression model including treatment, each subgroup factor, and their interaction term as covariates. $p_{interaction}$ values greater than or equal to 0·15 were considered not significant. Median overall survival estimates for each subgroup and by treatment are in the appendix (pp 25–26). HR was analysed using an unstratified Cox regression model including treatment as a covariate. ECOG=Eastern Cooperative Oncology Group. *Per case report form. †Other races include American Indian, Alaska Native, not reported, and unknown.

discontinuation due to other reason, whichever came first) was 16·3 months (IQR 8·6–21·8) in the tislelizumab group and 9·8 months (5·8–19·0) in the placebo group. Median study treatment exposure time was 6·4 months (IQR 3·3–11·1) in the tislelizumab group versus 4·9 months (2·5–8·3) in the placebo group. Exposure to different chemotherapy doublets among patients who received at least one dose of study treatment was similar between treatment groups (appendix p 20).

Overall, 38 (12%) of 326 patients in the tislelizumab group and 15 (5%) of 323 in the placebo group remained on study treatment as of data cutoff. 157 (48%) of 326 patients in the tislelizumab group and 177 (55%) of 323 in the placebo group were given subsequent systemic anticancer therapies after study treatment discontinuation; subsequent immunotherapy was given to 46 (14%) patients and 72 (22%) patients, respectively.

At the interim analysis, 422 deaths were observed: 196 (60%) of 326 patients in the tislelizumab group and 226 (70%) of 323 patients in the placebo group died. Median overall survival was 17·2 months (95% CI 15·8–20·1) in the tislelizumab group versus 10·6 months (95% CI 9·3–12·1) in the placebo group (stratified HR 0·66 [95% CI 0·54–0·80]; one-sided p<0·0001; figure 2A). Prespecified multivariable analysis adjusting for baseline PD-L1 status by the prespecified TAP score cutoff of 10% and other key baseline characteristics and prognostic factors supported the primary analysis of

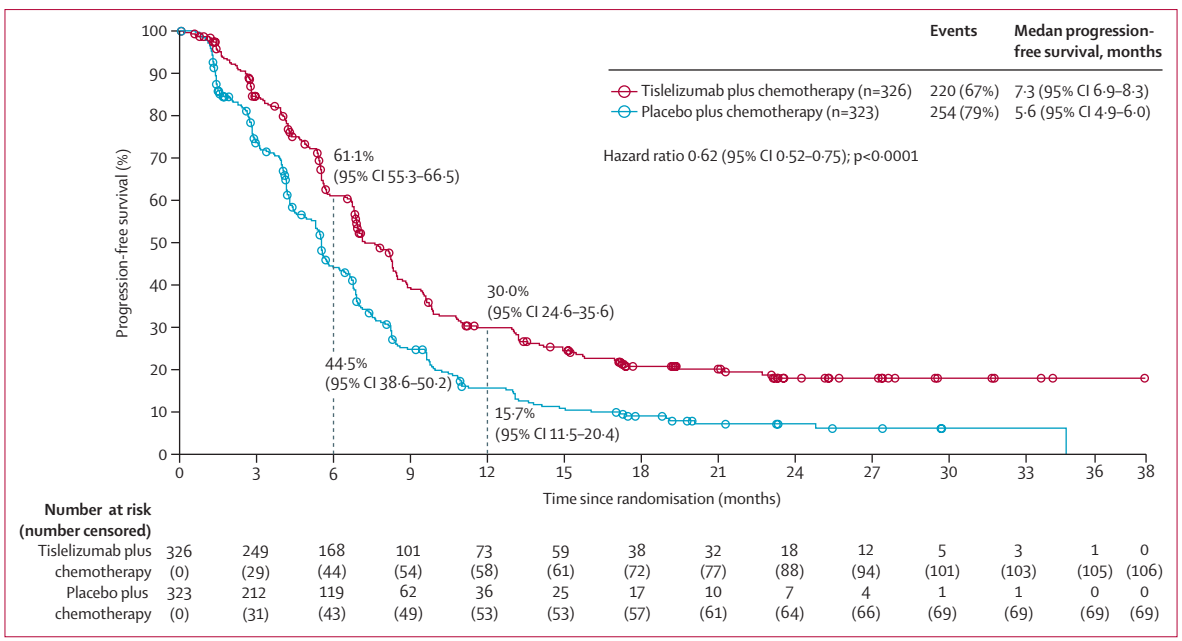

Figure 4: **Kaplan-Meier plot of progression-free survival by investigator assessment**

treatment effect on overall survival (appendix p 23). Overall survival by baseline PD-L1 TAP score subgroup is shown in figures 2B and C. Post-hoc interaction analysis between treatment group and baseline PD-L1 expression status (TAP score of ≥10%, <10%, or unknown) and PD-L1 subgroups defined by a PD-L1 combined positive score and tumour cell score showed no significant interaction between treatment effect and PD-L1 status (appendix p 24).

Prespecified patient subgroup analyses of overall survival are shown in figure 3. Post-hoc subgroup analyses across subgroups of chemotherapy doublet regimens are shown in the appendix (pp 25–26).

Significant improvement in investigator-assessed progression-free survival was observed with tislelizumab plus chemotherapy versus placebo plus chemotherapy (figure 4). The BIRC-assessed progression-free survival result was similar (appendix p 42). Progression-free survival across prespecified and post-hoc patient subgroups are shown in the appendix (p 27).

207 (63%) patients treated with tislelizumab plus chemotherapy versus 137 (42%) treated with placebo plus chemotherapy achieved an investigator-assessed objective response; odds ratio 2·38 (95% CI 1·73–3·27; p<0·0001; appendix p 28). Duration of response and disease control rate, per investigator assessment, are shown in the appendix (p 28). An improvement in overall response rate with tislelizumab versus placebo was seen in prespecified and post-hoc subgroup analyses (appendix p 29). BIRC-assessed objective response rate, duration of response, and disease control rate were similar to those assessed by the investigator (appendix p 30).

313 (97%) of 324 patients in the tislelizumab group and 309 (96%) of 321 patients in the placebo group had at least one treatment-related treatment-emergent adverse event; the most common treatment-related treatment-emergent adverse events are summarised in the appendix (pp 31–32). 216 (67%) patients in the tislelizumab group and 207 (64%) in the placebo group had a grade 3 or worse treatment-related treatment-emergent adverse event. The most common grade 3 or 4 treatment-related treatment-emergent adverse events were decreased neutrophil count (99 [31%] in the tislelizumab group *vs* 105 [33%] in the placebo group), decreased white blood cell count (35 [11%] *vs* 50 [16%]), and anaemia (47 [15%] *vs* 41 [13%]; table 2; appendix pp 33–38). A higher proportion of patients given tislelizumab plus chemotherapy than given placebo plus chemotherapy had serious treatment-related treatment-emergent adverse events (93 [29%] of 324 *vs* 62 [19%] of 321). The most common serious treatment-related treatment-emergent adverse events in the tislelizumab group versus the placebo group were pneumonia (seven [2%] of 324 *vs* eight [2%] of 321) and neutropenia (one [<1%] *vs* seven [2%]). 94 (29%) patients in the tislelizumab group versus 60 (19%) in the placebo group had treatment-related treatment-emergent adverse events that led to treatment discontinuation (appendix pp 39–40), and 226 (70%) versus 201 (63%), respectively, had treatment-related treatment-emergent adverse events that led to dose modification. Treatment-emergent adverse events leading to death occurred in 17 (5%) of 324 patients who received tislelizumab and 17 (5%) of 321 patients who received placebo (appendix p 41), of which six deaths in the tislelizumab group (gastrointestinal and upper gastrointestinal haemorrhage [n=2], myocarditis [n=1], pulmonary tuberculosis [n=1], electrolyte imbalance [n=1], and respiratory failure [n=1]) and

	Tislelizumab plus chemotherapy group (n=324)				Placebo plus chemotherapy group (n=321)			
	Grade 1–2	Grade 3	Grade 4	Grade 5*	Grade 1–2	Grade 3	Grade 4	Grade 5*
Any event	97 (30%)	153 (47%)	56 (17%)	7 (2%)	102 (32%)	148 (46%)	53 (17%)	6 (2%)
Anaemia	126 (39%)	46 (14%)	1 (<1%)	0	114 (36%)	41 (13%)	0	0
Decreased white blood cell count	108 (33%)	31 (10%)	4 (1%)	0	107 (33%)	45 (14%)	5 (2%)	0
Decreased appetite	107 (33%)	9 (3%)	0	0	108 (34%)	7 (2%)	0	0
Nausea	104 (32%)	8 (2%)	0	0	125 (39%)	5 (2%)	0	0
Peripheral sensory neuropathy	63 (19%)	10 (3%)	0	0	54 (17%)	7 (2%)	0	0
Alopecia	58 (18%)	0	0	0	63 (20%)	0	0	0
Diarrhoea	54 (17%)	9 (3%)	0	0	54 (17%)	5 (2%)	0	0
Decreased neutrophil count	54 (17%)	72 (22%)	27 (8%)	0	47 (15%)	70 (22%)	35 (11%)	0
Vomiting	53 (16%)	4 (1%)	0	0	67 (21%)	6 (2%)	1 (<1%)	0
Decreased platelet count	51 (16%)	8 (2%)	1 (<1%)	0	51 (16%)	3 (1%)	0	0
Stomatitis	45 (14%)	10 (3%)	3 (1%)	0	40 (12%)	7 (2%)	0	0
Decreased weight	45 (14%)	1 (<1%)	0	0	45 (14%)	0	0	0
Increased blood creatinine	42 (13%)	1 (<1%)	0	0	27 (8%)	1 (<1%)	0	0
Constipation	42 (13%)	0	0	0	40 (12%)	1 (<1%)	0	0
Increased aspartate aminotransferase	37 (11%)	4 (1%)	1 (<1%)	0	27 (8%)	1 (<1%)	1 (<1%)	0
Increased alanine aminotransferase	36 (11%)	5 (2%)	0	0	28 (9%)	4 (1%)	1 (<1%)	0
Hypoalbuminaemia	36 (11%)	0	0	0	25 (8%)	0	0	0
Fatigue	35 (11%)	13 (4%)	0	0	45 (14%)	8 (2%)	0	0
Malaise	35 (11%)	5 (2%)	1 (<1%)	0	47 (15%)	3 (1%)	0	0
Pruritis	34 (10%)	0	0	0	19 (6%)	0	0	0
Asthenia	33 (10%)	4 (1%)	0	0	38 (12%)	1 (<1%)	0	0
Hypoaesthesia	33 (10%)	1 (<1%)	0	0	39 (12%)	1 (<1%)	0	0
Hypothyroidism	31 (10%)	0	0	0	14 (4%)	0	0	0
Neutropaenia	29 (9%)	16 (5%)	7 (2%)	0	15 (5%)	19 (6%)	12 (4%)	0
Hypokalaemia	22 (7%)	17 (5%)	1 (<1%)	0	15 (5%)	7 (2%)	2 (1%)	0
Hyponatraemia	19 (6%)	17 (5%)	5 (2%)	0	23 (7%)	9 (3%)	1 (<1%)	0

Data are n (%). Data are shown for grade 1–2 treatment-related adverse events with an incidence of ≥10% and grade 3–5 treatment-related adverse events occurring in ≥5% of patients; all treatment-related adverse events of grade 3–5 are shown in the appendix (p 33). For each row category, a patient with two or more adverse events in that category was counted only once at the worst severity. *The grade 5 treatment-related adverse events included deaths due to disease progression. After excluding deaths due to disease progression, there were six events in the tislelizumab group and four events in the placebo group.

Table 2: **Treatment-related treatment-emergent adverse events (safety population)**

four deaths in the placebo group (pneumonia [n=1], septic shock [n=1], and unspecified death [n=2]) were determined to be treatment-related (table 2).

Discussion

In this global, phase 3 study in patients with advanced or metastatic oesophageal squamous cell carcinoma, the primary endpoint of overall survival was met at interim analysis. First-line treatment with tislelizumab plus chemotherapy met the criteria for superiority in terms of overall survival compared with placebo plus chemotherapy, with a stratified HR of 0·66 and 6·6-month improvement in median overall survival. Overall survival benefit was observed across all prespecified patient subgroups, including PD-L1 expression, investigator-chosen chemotherapy, geographical region, and race, with significantly longer investigator-assessed progression-free survival and a significantly greater investigator-assessed objective response rate in patients assigned to tislelizumab plus chemotherapy versus those assigned to placebo plus chemotherapy. Additionally, treatment with tislelizumab plus chemotherapy showed more durable responses versus placebo plus chemotherapy.

RATIONALE-306 enrolled patients regardless of PD-L1 expression status. The cutoff of 10% used for the prespecified TAP score was based on data from early phase studies of tislelizumab in oesophageal squamous cell carcinoma, and its clinical performance was assessed in the phase 3 RATIONALE-302 study of tislelizumab monotherapy versus chemotherapy in second-line treatment of oesophageal squamous cell carcinoma.[22] In RATIONALE-306, tislelizumab plus chemotherapy showed a significant and clinically meaningful improvement in overall survival versus placebo plus chemotherapy in patients with a PD-L1 TAP score of at least 10%, and a favourable but non-significant treatment effect in overall survival in patients with a PD-L1 TAP score of less than 10% in a prespecified exploratory

subgroup analysis (stratified HR 0·77). Although the overall survival benefit of tislelizumab plus chemotherapy appeared to be greater in patients with a PD-L1 TAP score of at least 10% than in those with a score of less than 10%, a post-hoc interaction analysis between treatment and baseline PD-L1 expression status by TAP score showed no significant difference in overall survival benefit for both groups on the basis of PD-L1 status. Similarly, the progression-free survival and objective response rate benefit of tislelizumab plus chemotherapy compared with placebo plus chemotherapy was observed in patients regardless of PD-L1 TAP score.

Increasing evidence suggests cytotoxic agents are substantially diverse in their immunostimulatory or regulatory properties, underscoring the importance of understanding ideal partnering of such agents with anti-PD-1 therapy.[12–14] No global consensus exists regarding optimal first-line chemotherapy for patients with advanced or metastatic oesophageal squamous cell carcinoma; some geographical regions prefer platinum with fluoropyrimidine and others prefer platinum with paclitaxel.[19] Preclinical data suggest fluoropyrimidine (an antimetabolite) and paclitaxel (an antimicrotubule agent) have differential effects on dendritic cell maturation and elimination of myeloid-derived suppressor cells in the tumour microenvironment, which might affect T-cell-dependent anti-tumour responses.[15,16,26] Most previous clinical studies in oesophageal squamous cell carcinoma examining anti-PD-1 therapy in combination with chemotherapy allowed only one platinum doublet (cisplatin plus fluorouracil[7,8] or cisplatin plus paclitaxel[9,11]), the exception being an almost entirely Asian trial (ORIENT-15), in which 93% of patients were given cisplatin plus paclitaxel and only 7% were given cisplatin plus fluorouracil,[10] thus restricting the comparison of fluoropyrimidine and paclitaxel.

To our knowledge, RATIONALE-306 is the first global study designed to investigate the effects of an anti-PD-1 monoclonal antibody in combination with different choices of chemotherapy regimen, choices that were largely based on regional preferences and the investigators' discretion. Platinum plus either fluoropyrimidine or paclitaxel is recommended by international, local, or country-specific treatment guidelines; as such, RATIONALE-306 has a global design that is closely aligned with real-world medical practice.[4] Random assignment to tislelizumab or placebo was stratified by investigators' choice of platinum with fluoropyrimidine versus platinum with paclitaxel, enabling unbiased analysis of tislelizumab efficacy versus placebo with these regimens; although, notably, RATIONALE-306 was not designed to identify the ideal chemotherapy partner for tislelizumab. In this study, an overall survival benefit of adding tislelizumab to chemotherapy was observed in both patients who received fluoropyrimidine and those who received paclitaxel, providing the first global evidence from a randomised controlled trial supporting combination of immune checkpoint inhibitors with taxane-containing therapy in patients with oesophageal squamous cell carcinoma. An improvement in overall survival favouring tislelizumab plus chemotherapy was also observed across both platinum-containing regimens, and specifically across each individual chemotherapy when further divided by cisplatin or oxaliplatin use in post-hoc analyses. Although the sample size of the oxaliplatin plus fluoropyrimidine subgroup was small, to our knowledge, this is the first randomised clinical trial to investigate clinical efficacy of anti-PD-1 therapy in combination with oxaliplatin plus fluoropyrimidine in oesophageal squamous cell carcinoma, and to show that the benefit is consistent with that observed with other anti-PD-1 plus chemotherapy combinations. These findings might provide clinically relevant insights, because oxaliplatin is a preferred platinum agent for treatment of oesophageal squamous cell carcinoma in the USA and other regions worldwide. In the present study, investigators chose oxaliplatin-based doublets for 55% of patients in countries outside of Asia, whereas 96% of patients in Asian countries received cisplatin-based doublets. Overall, our new data provide prospective clinical trial evidence for efficacy of tislelizumab plus different platinum-containing regimens, expanding the understanding of anti-PD-1 monoclonal antibody combination therapies worldwide.

The overall survival benefit with tislelizumab plus chemotherapy as a first-line treatment for advanced or metastatic oesophageal squamous cell carcinoma we found in RATIONALE-306 is similar to that from studies investigating other first-line anti-PD-1 monoclonal antibodies in populations with oesophageal squamous cell carcinoma.[7–11] Evidence for tislelizumab is noteworthy, as a larger proportion of patients in the chemotherapy (control) group compared with the tislelizumab group received subsequent immunotherapy treatment. Most phase 3 trials for anti-PD-1 monoclonal antibodies plus chemotherapy in first-line treatment of oesophageal squamous cell carcinoma enrolled almost entirely Asian populations.[9–11] KEYNOTE-590 and CheckMate 648 are the only two previous global studies investigating efficacy of anti-PD-1 monoclonal antibodies, pembrolizumab and nivolumab, respectively, in combination with chemotherapy (fluorouracil plus cisplatin in both trials), versus placebo plus chemotherapy or chemotherapy alone as a first-line treatment in patients with advanced or metastatic oesophageal squamous cell carcinoma.[7,8] Patients in countries outside of Asia comprised 48% of the KEYNOTE-590 cohort (included patients with adenocarcinoma and squamous cell histologies) and 30% of the CheckMate 648 cohort.[7,8] RATIONALE-306 was also designed as a global study to examine tislelizumab plus chemotherapy applicability to a diverse population; 163 (25%) of 649 participants from Europe, North America, and Oceania, and 486 (75%) from Asian countries were recruited to reflect the global distribution of oesophageal squamous cell carcinoma (with about 80%

of cases occurring in Asia).[27] Subgroup analyses from RATIONALE-306 showed overall survival benefit with tislelizumab plus chemotherapy versus placebo plus chemotherapy was consistent across Asia and other regions. Despite observed regional differences in epidemiology and clinical characteristics of oesophageal squamous cell carcinoma,[17–19] the observed tislelizumab treatment benefit for oesophageal squamous cell carcinoma was not affected in combination with chemotherapy (the current study) or as monotherapy (RATIONALE-302 study).[22] Similar observations were noted in CheckMate 648, in which overall survival benefit with nivolumab plus chemotherapy versus chemotherapy alone was seen in patients with oesophageal squamous cell carcinoma including both Asian and non-Asian populations (HR 0·74 [99·1% CI 0·58–0·96]).[8] In KEYNOTE-590, overall survival HR for pembrolizumab plus chemotherapy versus placebo plus chemotherapy was 0·63 (95% CI 0·49–0·80) in patients with oesophageal squamous cell carcinoma from Asia and 0·93 (95% CI 0·67–1·28) in those from other regions.[28] Additionally, overall survival benefit with tislelizumab plus chemotherapy compared with placebo plus chemotherapy was observed in both female and male patients in RATIONALE-306. Sex has been reported to affect immune function and differentially affect immune checkpoint inhibitors in some studies;[8,29,30] although these results should be interpreted with caution, given the small sample size of female patients, and further investigation is warranted.

The safety profile of tislelizumab plus chemotherapy was manageable. The proportion of patients having any grade and grade 3 or worse treatment-related treatment-emergent adverse events was similar between treatment groups. However, a greater proportion of patients treated with tislelizumab plus chemotherapy versus placebo plus chemotherapy had serious treatment-related treatment-emergent adverse events or treatment-related treatment-emergent adverse events leading to treatment discontinuation; additionally, median exposure duration was longer in the tislelizumab group than in the placebo group. The most common treatment-related treatment-emergent adverse events (all grade and grade 3 or 4) in both groups were consistent with known adverse events of the individual study treatment components, with no new safety signals identified. The trend in incidence of treatment-related diarrhoea, hypothyroidism, and pneumonitis (usually considered events of interest for anti-PD-1 therapies) between the tislelizumab group and the placebo group was similar to that observed in KEYNOTE-590 and CheckMate 648.[7,8] The safety profile of tislelizumab plus chemotherapy was generally similar to other anti-PD-1 monoclonal antibodies given with chemotherapy as a first-line treatment in patients with oesophageal squamous cell carcinoma.[7–11]

A limitation of this study is that PD-L1 status at baseline was not a prespecified stratification factor, and PD-L1 expression data were missing for 16% of patients. We also acknowledge challenges in data interpretation across studies when multiple PD-L1 scoring methods are used. Similar treatment benefit was observed with tislelizumab plus chemotherapy when PD-L1 expression status was defined by prespecified TAP score and in post-hoc assessment of combined positive and tumour cell scores. Future studies should further investigate concordance between PD-L1 assays in oesophageal squamous cell carcinoma and their ability to predict treatment efficacy with anti-PD-1 monoclonal antibodies and identify novel biomarkers through further investigation of the molecular and immune biology of oesophageal squamous cell carcinoma to better customise therapies.

First-line treatment of advanced or metastatic oesophageal squamous cell carcinoma with tislelizumab plus chemotherapy resulted in significant and clinically meaningfully overall survival benefit versus placebo plus chemotherapy. The safety profile of tislelizumab in combination with chemotherapy was manageable, with no new safety signals identified.

Contributors

JX, KK, ER, RAH, and HHY contributed to the conception and design of the study, acquisition of the data, and interpretation of the data. TK contributed to the conception and design of the study, acquisition of data, interpretation of data, and had additional study involvement. LW and LL contributed to the conception and design of the study, data analysis, and data interpretation. JS contributed to the design and conception of the study and data interpretation. LP, YJ, XW, YY, LS, EG, EVC, and PJ-F contributed to the data acquisition and data interpretation. YS, YP, SRP, JZ, RI, LW, and C-YL contributed to the data interpretation. HHY, LL, and LW had direct access to the data and contributed to the verification of the underlying data reported in the manuscript. All authors had access to all relevant aggregated study data, participated in writing or reviewing the manuscript, and provided final approval for the decision to submit the manuscript for publication.

Declaration of interests

KK has served in consulting roles for AstraZeneca, Bayer, BeiGene, Bristol Myers Squibb, MSD, and Ono Pharmaceutical and has received honoraria from Bristol Myers Squibb, MSD, and Ono Pharmaceutical. ER has participated on a data safety monitoring board or advisory board for BeiGene. RAH has served in consulting roles for Ipsen. SRP has received research grants from ImmuneOncia Therapeutics and Incyte and has received honoraria from MSD Korea. YY has received travel grants from BeiGene. LS has received grants from Baiji Shenzhou (Beijing) Biotechnology, Beijing Xiantong Biomedical Technology, Beihai Kangcheng (Beijing) Medical Technology, Jacobio Pharmaceuticals, Qilu Pharmaceutical, Yaojie Ankang (Nanjing) Technology, and Zai Lab Pharmaceutical (Shanghai); consulting fees from Haichuang Pharmaceutical, Harbour Biomed, and Mingji Biopharmaceutical; honoraria from CStone Pharmaceuticals, Hengrui, Hutchison Whampoa, and Zai Lab; and participated on an advisory board for Boehringer Ingelheim, Bristol Myers Squibb, Merck, MSD, Roche, Sanofi, and Servier. TK has received grants from Amgen, BeiGene, Bristol Myers Squibb, Chugai Pharmaceutical, EPS Corporation, Merck Biopharma, MSD, Ono Pharmaceutical, Parexel International, Shionogi & Co, and Taiho Pharmaceutical; honoraria from Bristol Myers Squibb, Covidien Japan, MSD, Oncolys BioPharma, Ono Pharmaceutical, and Taiho Pharmaceutical; and participated on a data safety monitoring board or advisory board for Astellas Pharma, Bristol Myers Squibb, Merck Biopharma, MSD, and Oncolys BioPharma. EVC has received grants from Amgen, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Celgene, Ipsen, Lilly, MSD, Merck KGaA, Novartis, Roche, and Servier and honoraria from AbbVie, Array, Astellas, AstraZeneca, Bayer, BeiGene, Biocartis, Boehringer Ingelheim, Bristol Myers Squibb, Celgene, Daiichi, Halozyme, GSK, Helsinn, Incyte, Ipsen, Janssen Research, Lilly, MSD, Merck KGaA,

Mirati, Novartis, Pierre Fabre, Roche, Seattle Genetics, Servier, Sirtex, Terumo, Taiho Pharmaceutical, TRIGR, and Zymeworks. PJ-F has received honoraria from Bristol Myers Squibb, HRA Pharmaceutical, LEO Pharma, Lilly, MSD, Novartis, Rovi, and Sanofi and participated on a data safety monitoring board or advisory board for Amgen, Bristol Myers Squibb, HRA Pharmaceutical, Lilly, MSD, Mylan, and Novartis. LW and JS are employees of BeiGene and have no additional competing interests. LL is an employee of BeiGene with stock or stock options. HHY has received grants from BeiGene, Bristol Myers Squibb, CARsgen Therapeutics, MacroGenics, and Merck; honoraria from BeiGene; travel grants from BeiGene; and has participated on a data safety monitoring board or advisory board for ALX Oncology, Amgen, AstraZeneca, Astellas, BeiGene, Bristol Myers Squibb, MacroGenics, Merck, OncXerna, Zymeworks, and Novartis. All other authors declare no competing interests.

Data sharing

On request, and subject to specific criteria, conditions, and exceptions, BeiGene will provide access to individual de-identified participant data from BeiGene-sponsored global interventional clinical studies conducted for medicines for indications that have been approved or in programmes that have been terminated. BeiGene will also consider requests for the protocol, data dictionary, and statistical analysis plan. Data requests can be submitted to datadisclosure@beigene.com.

Acknowledgments

We thank the participants of the study and all the study staff for their contributions. Third-party medical writing assistance, under the direction of the authors, was provided by Lynda McEvoy, of Ashfield MedComms, an Inizio company, and was funded by BeiGene. Biostatistics support was provided by Hongqian Wu, an employee of BeiGene.

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